

Janzen Choi

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Mechanical, Mechatronic, and Software Engineer

Summary

I am a recent PhD graduate whose research focused on applying machine learning and optimisation techniques to model material behaviour. I have strong programming skills (e.g., Bash, C, C++, Java, JavaScript, MATLAB, Python, SQL) and proficiency with a range of software tools (e.g., CAD, FEM, Microsoft Suite). I have also worked 5 years as a tutor (primary, secondary, university), 5 years as a student researcher, 1 year as a DevOps engineer, and 6 months as a research aide.

Work History

Graduate Researcher

[Australian Nuclear Science and Technology Organisation](#) (Sydney, Australia); ~1200 employees

Developed models to predict material behaviour under extreme conditions

12/20 – 12/25 (5 years, part-time)

- Applied machine learning and optimisation techniques (e.g., genetic algorithms, surrogate modelling) to calibrate material models against experimental data
- Presented research findings at meetings and conferences to internal and external stakeholders
- Supported colleagues in their model development workflows
- Co-authored three first-author peer-reviewed publications on modelling behaviour of Alloy 617

Languages: C++, Java, JavaScript, MATLAB, Python

Software: [Cubit](#) [Coreform](#), [MOOSE](#), [NEML](#), [Paraview](#)

Technical Research Aide

[Argonne National Laboratory](#) (Illinois, USA); ~5000 employees

Assisted with projects sponsored by the department of energy, regarding the physics-based modelling of Alloy 617, 709, and 316H steel

07/24 – 10/24 (3 months, full-time)

06/23 – 09/23 (3 months, full-time)

- Built surrogate models using deep neural networks to reduce the evaluation of physics-based models from days to milliseconds
- Developed elastic-viscoplastic and crystal plasticity finite element models to simulate mesoscale material deformation
- Tested and contributed to [NEML](#), Argonne's open-source material modelling codebase
- Co-authored two peer-reviewed publications on the creep and tensile behaviour of alloys

Languages: C++, MATLAB, Python

Software: [Cubit](#) [Coreform](#), [MOOSE](#), [NEML](#), [Paraview](#)

DevOps Engineer

[Mission Systems](#) (Sydney, Australia); ~30 employees

Developed software to streamline the software development process

11/22 – 11/23 (12 months, part-time)

- Designed and built a physical inventory tracking system from scratch, deployed in-house
- Wrote automation scripts to streamline recurring operations tasks

- Upgraded in-house codebase from Ubuntu 18.04 to 20.04
- Helped build vehicle simulations in Blender to support internal workflows
- Maintained Jenkins pipelines and used Docker to support CI/CD

Languages: Bash, C++, Python

Software: [Blender](#), [Docker](#), [Jenkins](#)

Course Demonstrator

[University of New South Wales](#) (Sydney, Australia); ~8000 employees

Delivered labs, tutorials, and marking support across mechatronics, mathematics, and design engineering courses

01/20 – 11/21 (22 months, part-time)

- Ran lab sessions and tutorials for MTRN4010 (Advanced Robotics), MATH2089 (Numerical Methods & Statistics), and DESN1000 (Introduction to Engineering Design)
- Supported classes of up to 30 students, assisting with practical exercises and course material
- Marked assignments and provided feedback to students

Languages: C++, MATLAB, Python

Software: [Microsoft Suite](#), [Webots](#)

Education and Awards

2026	PhD (Mechanical Engineering)	UNSW Sydney
2022	Bachelor's Degree (Mechatronics), 92HD	UNSW Sydney
2022	Bachelor's Degree (Computer Science), 92HD	UNSW Sydney
2026	<u>University Medal</u> (Mechatronics)	UNSW Sydney
2022	<u>RS Components Prize</u> (Mechatronics)	UNSW Sydney
2021	<u>Dean's Award</u> (Mechatronics)	UNSW Sydney
2020	<u>Dean's Award</u> (Computer Science)	UNSW Sydney

Publications

2026	<i>Multi-Objective Surrogate-Assisted Calibration of CPFEM Models Using Macroscopic Response and In Situ EBSD Measurements of Grain Reorientation Trajectories</i> (Acta Materialia) (link)
2025	<i>Multi-Objective Calibration of Elastic-Viscoplastic Models to capture the Elevated-Temperature Creep and Tensile Behaviour of Alloy 617</i> (International Journal of Pressure Vessels and Piping) (link)
2023	<i>A mechanistic model for creep and thermal aging in Alloy 709</i> (U.S. Department of Energy Office of Scientific and Technical Information) (link)
2022	<i>On the prediction of creep behaviour of Alloy 617 using Kachanov-Rabotnov model coupled with multi-objective genetic algorithm optimisation</i> , (International Journal of Pressure Vessels and Piping) (link)
2021	<i>Multi-objective genetic algorithms for material parameter optimisation to predict high-temperature creep behaviour</i> (British Society for Strain Measurement) (link)

Projects

Project Prostheses

Independent Student Project (10 members)

Designed modular, low-cost prostheses with interchangeable modules to accommodate different activities and physical growth

09/21 – 09/22 (12 months)

- Focused on research, model development (CAD), and testing of the knee-joint module in lower-limb prostheses
 - Produced reports and presentations for external stakeholders
 - Contributed to a working prototype through 3D printing
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AI 4 Everyone

[ChallENG Vertically Integrated Project](#) (~30 members)

Conducted research on robotics, computer vision, and disaster response via underground autonomous vehicles; founding member of UNSW's RoboCup@Rescue team

12/20 – 12/21 (12 months)

- Composed comprehensive literature reviews on underground, aerial, and marine UAVs and their use across disaster scenarios
 - Helped develop a SLAM-based navigation system using ROS and MATLAB
 - Tested computer vision models (YOLO) via simulation as part of the computer vision sub-team
 - Authored foundational documentation to guide future cohorts
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AIAA Design, Build, Fly

[ChallENG Aircraft Design Competition](#) (~50 members)

Participated in a university project for designing a competition remote-control aircraft tasked to transport payload figurines

08/20 – 06/21 (14 months)

- Helped design and manufacture the mechanism for grabbing and securing the payload
 - Collaborated within a large, multi-disciplinary team spanning aerodynamics, structures, and electronics, coordinating across sub-teams to integrate components
 - Authored technical reports and presentations to satisfy competition documentation and stakeholder requirements
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Coder One AI Sports Challenge

[AI Bot Competition](#) (3 members)

Competed in a multiplayer, turn-based strategy game using a self-programmed AI agent; team reached the semi-finals (8/16 teams)

11/20 – 12/20 (1 month)

- Developed a reinforcement learning-based bot to compete against other teams' agents
 - Programmed the agent's decision-making logic in Python against Coder One's game engine
 - Designed objectives for navigation, hazard avoidance, and bomb placement
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TEDI-London

[Product Design Program](#) (7 members)

Participated in a program for designing a dementia-care product

04/20 – 08/20 (4 months)

- Designed mountable sensory (audio, visual, and tactile) tiles to support mood and engagement for people with dementia
- Organised and managed workflows across design, research, and reporting
- Authored reports and presentations to communicate decisions and progress to stakeholders
- Contributed to a working prototype through 3D printing and laser cutting